1 Solution — GIAM 7.2 Problem 8

1.1 Problem

Construct magic squares of order 4 and 5.

1.2 Definitions and the Magic Constant

A *magic square of order* n is an $n \times n$ array filled with the integers $1, 2, \ldots, n^2$ such that the sum along every row, every column, and both main diagonals equals the same value M_n, called the *magic constant*.

The value of M_n is forced by a counting argument. The grand total of all entries is

$$1 + 2 + \cdots + n^2 \;=\; \frac{n^2(n^2+1)}{2}.$$

This total is also n times the row sum (since the n rows partition the entries), so

$$M_n \;=\; \frac{1}{n} \cdot \frac{n^2(n^2+1)}{2} \;=\; \frac{n(n^2+1)}{2}.$$

For $n = 4$: $M_4 = 4 \cdot 17/2 = 34$. For $n = 5$: $M_5 = 5 \cdot 26/2 = 65$.

We construct each magic square by a *systematic method* rather than ad-hoc trial.

1.3 Order 4 — The Doubly-Even (Diagonal-Swap) Method

When n is divisible by 4 (i.e., *doubly even*), the following procedure produces a magic square. We illustrate for $n = 4$.

Step 1 — Fill in row-major order. Place 1 through 16 left-to-right, top-to-bottom:

```
 1   2   3   4
 5   6   7   8
 9  10  11  12
13  14  15  16
```

Step 2 — Mark cells on the two main diagonals. The main diagonal is $(1,1),(2,2),(3,3),(4,4)$ and the anti-diagonal is $(1,4),(2,3),(3,2),(4,1)$; together these are 8 cells (no overlaps for $n=4$). Marked values: $1,6,11,16,4,7,10,13$.

Step 3 — Complement the unmarked cells. Replace every *unmarked* value v by its "complement" $n^2+1-v=17-v$:

$$2\to 15,\quad 3\to 14,\quad 5\to 12,\quad 8\to 9,\quad 9\to 8,\quad 12\to 5,\quad 14\to 3,\quad 15\to 2.$$

Marked cells keep their original values. The resulting grid is

```
 1  15  14   4
12   6   7   9
 8  10  11   5
13   3   2  16
```

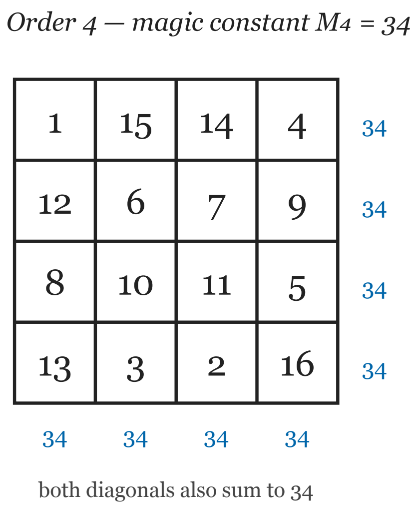

Figure 1.1: The constructed order-4 magic square. Every row, every column, and both diagonals sum to the magic constant $M_4 = 34$.

Verification ($M_4 = 34$):

Row 1	$1, 15, 14, 4$	34
Row 2	$12, 6, 7, 9$	34
Row 3	$8, 10, 11, 5$	34
Row 4	$13, 3, 2, 16$	34
Col 1	$1, 12, 8, 13$	34
Col 2	$15, 6, 10, 3$	34
Col 3	$14, 7, 11, 2$	34
Col 4	$4, 9, 5, 16$	34
Main diagonal ↘	$1, 6, 11, 16$	34
Anti-diagonal ↙	$4, 7, 10, 13$	34

Table 1.1

All ten lines sum to 34. ✓

Why does the method work? In the initial $1, 2, \ldots, 16$ grid, every row i sums to $4(4i-3)+6 = 16i-6$, which varies from row to row. The complementation trick balances the imbalance: in each row, the two marked cells (on the diagonals) have *exactly* the same sum as the two unmarked cells (a non-trivial coincidence of the natural numbering pattern). Replacing the unmarked values v by $17-v$ then makes each row sum to (marked sum) $+(34-$ unmarked sum$) = (marked sum) - (unmarked sum$) + 34 = 0 + 34 = 34. The same balance holds along columns and diagonals.

1.4 Order 5 — The Siamese Method (de la Loubère)

For *odd* n, the *Siamese method* — published by Simon de la Loubère in 1693 after his diplomatic mission to Siam — produces a magic square by walking the integers $1, 2, \ldots, n^2$ through the grid by a fixed rule.

Algorithm.

1. Place 1 in the middle of the top row, position $(1, \lceil n/2 \rceil)$. For $n = 5$, that's $(1, 3)$.
2. To place $k + 1$: from the current cell, try moving *one row up and one column right.* Wrap around (row 0 becomes row n; column $n + 1$ becomes column 1).
3. *If the target cell is already filled,* instead move *one row down* from the current cell.
4. Repeat for $k = 1, 2, \ldots, n^2 - 1$.

Trace for $n = 5$. Selected key steps (a "blocked" step occurs every n steps):

k	**Current**	**Up-right candidate**	**Action**	**Place**
1	—	—	start	1 at $(1, 3)$
2	$(1, 3)$	$(0, 4) \to (5, 4)$	wrap row	2 at $(5, 4)$
3	$(5, 4)$	$(4, 5)$	place	3 at $(4, 5)$
4	$(4, 5)$	$(3, 6) \to (3, 1)$	wrap col	4 at $(3, 1)$
5	$(3, 1)$	$(2, 2)$	place	5 at $(2, 2)$
6	$(2, 2)$	$(1, 3)$ filled	**go down**	6 at $(3, 2)$
7–10				continue diagonal
11	$(4, 1)$	$(3, 2)$ filled	**go down**	11 at $(5, 1)$
$\vdots$				
25	$(1, 2)$	$(0, 3) \to (5, 3)$	wrap row	25 at $(5, 3)$

Table 1.2

Filling out all 25 steps yields

```
17  24   1   8  15
23   5   7  14  16
 4   6  13  20  22
10  12  19  21   3
11  18  25   2   9
```

Order 5 — magic constant M5 = 65

17	24	1	8	15	65
23	5	7	14	16	65
4	6	13	20	22	65
10	12	19	21	3	65
11	18	25	2	9	65
65	65	65	65	65	

both diagonals also sum to 65

Figure 1.2: The Siamese construction yields this order-5 magic square. Every row, every column, and both diagonals sum to $M_5 = 65$.

Verification ($M_5 = 65$):

Line	**Entries**	**Sum**
Row 1	$17, 24, 1, 8, 15$	65
Row 2	$23, 5, 7, 14, 16$	65
Row 3	$4, 6, 13, 20, 22$	65
Row 4	$10, 12, 19, 21, 3$	65
Row 5	$11, 18, 25, 2, 9$	65
Col 1	$17, 23, 4, 10, 11$	65
Col 2	$24, 5, 6, 12, 18$	65
Col 3	$1, 7, 13, 19, 25$	65
Col 4	$8, 14, 20, 21, 2$	65
Col 5	$15, 16, 22, 3, 9$	65
Main diagonal $\searrow$	$17, 5, 13, 21, 9$	65

Anti-diagonal ↙	$15, 14, 13, 12, 11$	65

Table 1.3

All twelve lines sum to 65. ✓

(The anti-diagonal is particularly satisfying: $11, 12, 13, 14, 15$ — five consecutive integers centered on the middle value 13, which is itself the median of $\{1, \ldots, 25\}$.)

Why does the Siamese method work? Each entry k can be written $k = n \cdot a_k + b_k + 1$ where $a_k \in \{0, 1, \ldots, n-1\}$ encodes the "block" and $b_k \in \{0, 1, \ldots, n-1\}$ encodes the "offset" within the block. The Siamese walk arranges the values so that both the a-coordinates and the b-coordinates form a *Latin square* — that is, each row and each column contains each digit $\{0, 1, \ldots, n-1\}$ exactly once. The row sum is therefore $n \cdot (0 + 1 + \cdots + (n-1)) + (0 + 1 + \cdots + (n-1)) + n = n \cdot \frac{n(n-1)}{2} + \frac{n(n-1)}{2} + n = \frac{n(n^2+1)}{2} = M_n$, independent of the row. Columns and diagonals follow the same logic.

1.5 Remark — Dürer's Magic Square

A famous order-4 magic square appears in Albrecht Dürer's 1514 engraving *Melencolia I*:

```
16   3   2  13
 5  10  11   8
 9   6   7  12
 4  15  14   1
```

This is not the square produced by the doubly-even method above, but a *different* order-4 magic square (there are 880 essentially distinct order-4 magic squares). Dürer's has additional symmetries: every 2×2 block sums to 34, opposite cells across the center sum to 17, and crucially the two middle cells of the bottom row read "15 14" — the year of the engraving.

Magic squares of every order $n \geq 3$ exist:

- $n = 3$: a unique magic square (up to rotation and reflection), the *Lo Shu* square attributed to ancient Chinese mathematics. $M_3 = 15$.
- $n = 4$: 880 essentially distinct magic squares; construct via doubly-even method or by hand.
- $n = 5$: many; Siamese method gives one canonical example. $M_5 = 65$.
- $n = 6$: the *singly-even* case ($n \equiv 2 \pmod 4$); construction is the trickiest, often via *LUX method* or *Conway's method*.
- $n \geq 7$ odd: Siamese method always works.
- $n \geq 8$ doubly-even: doubly-even method generalizes.

The case $n = 2$ is the unique impossible case (a 2×2 magic square would need all four entries equal, which contradicts using each of $1, 2, 3, 4$ exactly once).

1.7 Remark — A Counting Cross-Check

The magic constant formula $M_n = n(n^2 + 1)/2$ can be re-derived from Problem 1-style counting. The total of all entries is the sum $\sum_{k=1}^{n^2} k = \binom{n^2+1}{2}$ — a fact that also appeared in Chapter 7.1 Problem 9 (handshake count). Dividing by n rows gives M_n. So magic squares are a place where the elementary counting of the previous section reappears in a striking visual form.